UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER SECURITY
COURSE CODE	:	CSC662
EXAMINATION	:	OCTOBER 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC662 mock paper (October 2026). Computed answers show working; theory answers are supported by lecture slides (not lecturer-marked) with a verbatim quote and slide citation.

2. Supported theory marking points quote the lecture decks; any unresolved slot would remain a visible placeholder.

3. Part A = 10 x (2 + 6); Part B = 2 x (5 + 5); total = 100 marks.

4. Computed: 0 slot(s). Supported theory: 24 slots. Placeholders: 0. Questions and answers are for local study use only.

5. Near-copy rejection used all 35 past papers; style widening is recorded per slot.

FOR MARKERS' USE ONLY

This marking scheme consists of 16 printed pages

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CSC662 - MARKING SCHEME (MOCK OCTOBER 2026)

Labels: computed / supported by lecture slides (not lecturer-marked) / placeholder. No official marking scheme exists for the real CSC662 papers; this is generated for the mock.

Entailment labels: supported = the quote directly supports the marking point (or the point passed the stem-overlap gate without an LLM check); partly supported = the quote supports part of the point or is ambiguous; not supported = the quote does not support the point. Only 'not supported' points triggered regeneration.

Part A: 10 x (2 + 6) = 80 marks. Part B: 2 x (5 + 5) = 20 marks. Total: 100 marks. Slots: 0 computed, 24 supported by lecture slides, 0 placeholder.

Methods: computed answers use verified solvers. Theory answers are supported by lecture slides (not lecturer-marked). Each marking point carries a verbatim quote, a deck/slide/chunk citation and a term-overlap score; low overlap is flagged for human review. Sources are restricted to the slot topic's allowed decks, and a slide is not reused across questions where an alternative exists.

PART A - QUESTION 1

A1a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

State TWO (2) characteristics of a Trojan horse as a type of malware.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Trojans disguise themselves as useful or benign software in order to be run by the user.	"They usually masquerade as helpful or harmless programs"	051 - Software Security, slide 12 051 - Software Security::s12::1	supported	0.00 LOW
2	Once active, a Trojan steals private data, deletes files, logs keystrokes or opens a way for additional malware.	"begin to steal private data, delete files, log keystrokes or provide an opening for further malware to get installed"	051 - Software Security, slide 12 051 - Software Security::s12::1	supported (stem gate)	0.64

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 051 - Software Security, slide 12

A1b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain six (6) techniques by which malware infects host programs or propagates across systems, as described in the lecture material on software security.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	A virus can infect a COM executable by appending its virus body to the host program.	"Append a virus body to a program"	051 - Software Security, slide 9 051 - Software Security::s09::1	supported	0.44 LOW
2	The virus saves the original program entry point inside its own virus body before taking control.	"Save an entry point to a program in a virus body"	051 - Software Security, slide 9 051 - Software Security::s09::1	supported (stem gate)	0.50
3	The program entry point is replaced with a jump into the virus body so the virus executes first.	"Replace a program entry point with a jump to a virus body"	051 - Software Security, slide 9 051 - Software Security::s09::1	supported (stem gate)	0.70
4	For EXE files, the virus overwrites the program header so that the entry point is switched to the virus.	"Over write a program header to switch the entry point to a virus"	051 - Software Security, slide 10 051 - Software Security::s10::1	supported (stem gate)	0.67
5	The Morris worm propagated across the Arpanet by exploiting a debugging feature of sendmail that allowed remote execution.	"Used a debugging feature of sendmail (remote execution)"	051 - Software Security, slide 7 051 - Software Security::s07::1	partly supported	0.36 LOW

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No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
6	The Brain virus spread in the wild by being included in the boot sector of distributed software diskettes.	"Included in a boot sector of distributed software diskettes"	051 - Software Security, slide 7 051 - Software Security::s07::1	supported (stem gate)	0.60

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 051 - Software Security, slide 10; 051 - Software Security, slide 7; 051 - Software Security, slide 9

PART A - QUESTION 2

A2a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

List two ways in which morals, ethics and laws differ from one another as forms of guidance for conduct.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Morals are described as traditions of belief concerning right and wrong conduct within a society.	"Morals are traditions of belief about right and wrong conduct"	071 - Ethics and Computer Security, slide 5 071 - Ethics and Computer Security::s05::1	supported (stem gate)	0.67
2	Laws are formal rules of conduct imposed by a sovereign authority such as a government on its citizens.	"Laws are formal rules of conduct that a sovereign authority, such as a government, imposes on its subjects or citizens"	071 - Ethics and Computer Security, slide 5 071 - Ethics and Computer Security::s05::1	supported (stem gate)	1.00

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 071 - Ethics and Computer Security, slide 5

A2b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Refer to your answer in (a). Discuss how a professional code of ethics and conduct guides a computing professional's behaviour, using the ACM code as your example. Your discussion should cover six distinct requirements of such a code.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	State that a code of ethics is a documented set of acceptable behaviours for a professional group, defined by that group.	"Documented set of acceptable behaviors for professional or social groups."	071 - Ethics and Computer Security, slide 6 071 - Ethics and Computer Security::s06::1	supported (stem gate)	0.56
2	Note that the ACM is a professional body that has established a code of ethics and professional conduct.	"Association of Computing Machinery ACM (founded 1947) has a code of ethics and professional conduct."	071 - Ethics and Computer Security, slide 7 071 - Ethics and Computer Security::s07::1	supported (stem gate)	0.67
3	Explain that the code obliges the professional to contribute to society and human well-being, recognising all people as stakeholders.	"Contribute to society and to human well-being, acknowledging that all people are stakeholders in computing."	071 - Ethics and Computer Security, slide 11 071 - Ethics and Computer Security::s11::1	supported (stem gate)	0.64

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No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
4	Explain that the code requires the professional to respect privacy and to honour confidentiality.	"Respect privacy."	071 - Ethics and Computer Security, slide 11 071 - Ethics and Computer Security::s11::1	partly supported	0.29 LOW
5	Explain that the code requires the professional to strive for high quality in the processes and products of professional work.	"Strive to achieve high quality in both the processes and products of professional work."	071 - Ethics and Computer Security, slide 12 071 - Ethics and Computer Security::s12::1	supported (stem gate)	0.78
6	Explain that the code requires the professional to give thorough evaluations of computer systems and their impacts, including possible risks.	"Give comprehensive and thorough evaluations of computer systems and their impacts, including analysis of possible risks."	071 - Ethics and Computer Security, slide 12 071 - Ethics and Computer Security::s12::1	supported (stem gate)	0.73

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 071 - Ethics and Computer Security, slide 11; 071 - Ethics and Computer Security, slide 12; 071 - Ethics and Computer Security, slide 6; 071 - Ethics and Computer Security, slide 7

PART A - QUESTION 3

A3a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain the Chinese Wall model as a formal security model, stating the concern it is designed to address and how its access decisions differ from static policies.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	The Chinese Wall policy is intended to address conflicts of interest affecting a consultant or contractor.	"Brewer and Nash's Chinese Wall Policy is designed to address a very specific concern: conflicts of interest by a consultant or contractor."	024 - Formal Security Model, slide 27 024 - Formal Security Model::s27::1	supported (stem gate)	0.80
2	Access rights under this policy are not static but depend on the subject's history of past accesses.	"Notice that permissions change dynamically. The access rights that any subject enjoys depends on the history of past accesses"	024 - Formal Security Model, slide 24 024 - Formal Security Model::s24::1	supported (stem gate)	0.55

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 024 - Formal Security Model, slide 24; 024 - Formal Security Model, slide 27

A3b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Describe six aspects of access control structures, covering the access control matrix, access control lists and capabilities, including the advantages and disadvantages of the matrix and of ACLs.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	State that an access control matrix is a fundamental access control structure.	"ACM is a basic control structure."	023 - Access Control and Authorization, slide 8 023 - Access Control and Authorization::s08::1	supported	0.40 LOW

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No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
2	Give clarity of definition as an advantage of the access control matrix.	"Advantages: clarity of definition easy to verify"	023 - Access Control and Authorization, slide 8 023 - Access Control and Authorization::s08::1	supported	0.33 LOW
3	Give poor scalability as a disadvantage of the access control matrix.	"Disadvantages: poor scalability poor handling of changes"	023 - Access Control and Authorization, slide 8 023 - Access Control and Authorization::s08::1	supported (stem gate)	0.50
4	Explain that an ACL describes access rights from the perspective of the object.	"ACL is an object-centered description of access rights:"	023 - Access Control and Authorization, slide 10 023 - Access Control and Authorization::s10::1	supported (stem gate)	0.80
5	Identify difficulty of revocation as a drawback of ACLs.	"difficulty of revocation difficulty of sharing"	023 - Access Control and Authorization, slide 10 023 - Access Control and Authorization::s10::1	supported	0.25 LOW
6	Name capabilities (C-List) as one of the structures used to implement access policies.	"access control matrix (ACM) Capabilities (C-List) access control lists (ACL)"	023 - Access Control and Authorization, slide 7 023 - Access Control and Authorization::s07::1	supported	0.40 LOW

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 023 - Access Control and Authorization, slide 10; 023 - Access Control and Authorization, slide 7; 023 - Access Control and Authorization, slide 8

PART A - QUESTION 4

A4a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain how the classes of threats described in the lecture affect the confidentiality, integrity and availability of information assets.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Disclosure threats such as snooping, sniffing and wiretapping breach confidentiality, which confidentiality services are meant to counter.	"Disclosure - unauthorized access to information"	011 - Introduction to Computer Security, slide 10 011 - Introduction to Computer Security::s10::1	partly supported	0.09 LOW
2	Deception and disruption threats involve modification of data, so integrity services are the countermeasure for these threats.	"Integrity services counter this threat"	011 - Introduction to Computer Security, slide 10 011 - Introduction to Computer Security::s10::1	partly supported	0.33 LOW

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Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 011 - Introduction to Computer Security, slide 10

A4b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

State SIX (6) facts about the Samy worm and the cross-site scripting (XSS) attack it exploited, covering how the malicious script was injected, how it spread to other users, and why unfiltered user input enables such attacks.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	The worm injected JavaScript into a user profile using document.body.innerHTML.	"Inject Javascript into a user profile using document.body.innerHTML"	052 - Web Security, slide 15 052 - Web Security::s15::1	supported (stem gate)	0.78
2	The worm obtained the list of the user's heroes by issuing a GET request on a user profile.	"Get the list of user's heros by using a GET on a user profile"	052 - Web Security, slide 15 052 - Web Security::s15::1	supported	0.38 LOW
3	The worm added Samy as a field by performing XML-HTTP on the addFriends page.	"Add Samy as a field by performing XML-HTTP on the addFriends page"	052 - Web Security, slide 15 052 - Web Security::s15::1	supported (stem gate)	0.78
4	Any user visiting an infected user's page had their profile and friends list infected with the JavaScript and with Samy.	"Any user who visits an infected user's page will gets profile infected with Javascript and his friends list infected with Samy"	052 - Web Security, slide 15 052 - Web Security::s15::1	supported (stem gate)	0.80
5	The infection eventually reached about 1,000,000 MySpace profiles.	"Eventually about 1,000,000 MySpace profiles were infected."	052 - Web Security, slide 15 052 - Web Security::s15::1	supported (stem gate)	0.71
6	Any web application that returns user input without filtering is a potential XSS vulnerability.	"Any web application that returns user input without filtering is a potential XSS vulnerability!"	052 - Web Security, slide 13 052 - Web Security::s13::1	supported (stem gate)	0.91

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 052 - Web Security, slide 13; 052 - Web Security, slide 15

PART A - QUESTION 5

A5a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain what a buffer overflow is and how it can be exploited, referring to the behaviour of strcpy() and the effect on the stack.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	strcpy() copies the input string without verifying its length, so a longer string than the destination buffer causes an overflow.	"strcpy() do not check the string length"	051 - Software Security, slide 23 051 - Software Security::s23::1	partly supported	0.23 LOW
2	Because no bound checking is performed, the overflowing buffer content spills into the adjacent higher stack area.	"Without proper bound checking a buffer content can overspill into adjacent upper stack area."	051 - Software Security, slide 21 051 - Software Security::s21::1	supported (stem gate)	0.61

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 051 - Software Security, slide 21; 051 - Software Security, slide 23

A5b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Describe six Windows security mechanisms and features, covering how user accounts and objects are managed, how access control is applied, how subjects obtain credentials, what Active Directory organises, how access rights are encoded, and why the complexity of these mechanisms can itself be a risk.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Windows includes elaborate mechanisms for managing user accounts and objects and for controlling access.	"Windows systems contain complex security mechanisms for user account and object management as well as access control."	041 - Windows Security, slide 21 041 - Windows Security::s21::1	supported (stem gate)	0.80
2	Discretionary access control lists allow fine-grained control over diverse entities both in Windows itself and in applications.	"DACS enable fine-grained access control of heterogeneous entities in Windows itself as well as applications."	041 - Windows Security, slide 21 041 - Windows Security::s21::1	supported (stem gate)	0.54
3	The very complexity of Windows security can backfire, since misconfiguration and implementation bugs can cause serious incidents.	"The complexity of Windows security mechanisms is also its enemy: misconfiguration as well as implementation bugs may lead to severe security incidents."	041 - Windows Security, slide 21 041 - Windows Security::s21::1	supported	0.46 LOW
4	In Windows, processes and threads act as the active subjects of the operating system.	"Windows subjects are processes and threads."	041 - Windows Security, slide 11 041 - Windows Security::s11::1	supported (stem gate)	0.50
5	A subject's security credentials are kept in a token, which maps principals to subjects and can carry extra security attributes.	"Security credentials for a subject are stored in a token."	041 - Windows Security, slide 11 041 - Windows Security::s11::1	partly supported	0.36 LOW
6	Active Directory, introduced with Windows 2000, is an LDAP-like directory service used to organise users and groups, credentials and certificates, resources, policies, DNS and trust.	"Active directory introduced in Windows 2000 is an LDAP-like directory service for organization of system resources:"	041 - Windows Security, slide 6 041 - Windows Security::s06::1	supported (stem gate)	0.53

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 041 - Windows Security, slide 11; 041 - Windows Security, slide 21; 041 - Windows Security, slide 6

PART A - QUESTION 6

A6a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

State the two stages of the biometric authentication process and, for each stage, state the activities it involves.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	The identification stage involves acquisition, creation of master characteristics and storage of master characteristics.	"Identification Stage Acquisition Creation of Master characteristics Storage of Master characteristics"	072 - Biometrics and Authentication, slide 5 072 - Biometrics and Authentication:: s05::1	supported (stem gate)	0.86
2	The authentication stage involves acquisition(s), comparison and decision.	"Authentication Stage Acquisition(s) Comparison Decision"	072 - Biometrics and Authentication, slide 5 072 - Biometrics and Authentication:: s05::1	supported (stem gate)	0.83

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.

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Citations: 072 - Biometrics and Authentication, slide 5

A6b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Discuss biometric technologies by categorising the types of biometric applications available today and, for each category, giving examples. Then discuss two disadvantages associated with specific biometric technologies.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Biometric applications are divided into two sectors, namely psychological and behavioural.	"Biometric applications available today are categorized into 2 sectors"	072 - Biometrics and Authentication, slide 4 072 - Biometrics and Authentication:: s04::1	supported (stem gate)	0.57
2	The psychological category includes iris, fingerprints, hand, retinal and face recognition.	"Psychological: Iris, Fingerprints, Hand, Retinal and Face recognition"	072 - Biometrics and Authentication, slide 4 072 - Biometrics and Authentication:: s04::1	supported (stem gate)	0.78
3	The behavioural category includes voice, typing pattern and signature.	"Behavioural: Voice, Typing pattern, Signature"	072 - Biometrics and Authentication, slide 4 072 - Biometrics and Authentication:: s04::1	supported (stem gate)	0.67
4	Fingerprint biometrics suffer from racial issues, dirt, grime and wounds, and placement of the finger.	"Racial issues Dirt , grime and wounds Placement of finger"	072 - Biometrics and Authentication, slide 7 072 - Biometrics and Authentication:: s07::1	supported (stem gate)	0.70
5	Fingerprints can be spoofed, so liveness detection is important.	"Can be spoofed - liveness important!"	072 - Biometrics and Authentication, slide 7 072 - Biometrics and Authentication:: s07::1	supported (stem gate)	0.67
6	Retinal scanning raises acceptability concerns relating to light exposure and hygiene.	"Acceptability concerns Light exposure Hygiene"	072 - Biometrics and Authentication, slide 9 072 - Biometrics and Authentication:: s09::1	supported (stem gate)	0.56

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 072 - Biometrics and Authentication, slide 4; 072 - Biometrics and Authentication, slide 7; 072 - Biometrics and Authentication, slide 9

PART A - QUESTION 7

A7a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

State TWO (2) limitations of misuse detection systems when used to detect attacks.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap

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No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Misuse detection systems are ineffective against previously unknown attack methods.	"They can detect many or all known attack patterns, but they are of little use for unknown attack methods."	062 - Intrusion Detection System, slide 17 062 - Intrusion Detection System::s17::1	supported	0.44 LOW
2	State transition analysis cannot detect denial of service attacks.	"They can't detect denail of service attacks."	062 - Intrusion Detection System, slide 21 062 - Intrusion Detection System::s21::1	supported	0.43 LOW

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 062 - Intrusion Detection System, slide 17; 062 - Intrusion Detection System, slide 21

A7b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

A security analyst is preparing a briefing on malware threat classes and the countermeasures that address them. Explain the threat posed by rootkits, spyware and adware, and the countermeasures or defensive considerations associated with each.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Rootkits conceal malicious processes and software on the infected system so that privileged remote access can be maintained.	"Rootkits are a type of malware that mask malicious processes and software in the infected systems to ensure privileged remote access."	051 - Software Security, slide 13 051 - Software Security::s13::1	supported (stem gate)	0.75
2	Rootkits are typically installed after a vulnerability has been found and exploited, keeping the machine under the attacker's control.	"They are usually installed after a vulnerability has been found and detected so that the infected computer continues to be under the control of the malware author."	051 - Software Security, slide 13 051 - Software Security::s13::1	supported (stem gate)	0.54
3	Detecting rootkits is difficult because they are designed to hide from the very detection software used against them.	"Rootkit detection is quite difficult as they are designed to stay hidden from the very software used to detect them."	051 - Software Security, slide 13 051 - Software Security::s13::1	supported (stem gate)	0.73
4	Spyware steals user credentials and records activity such as sites visited and keystrokes, and can redirect the browser to unwanted sites.	"They steal user logins and passwords, document sites visited, log keystrokes and even redirect browsers to unwanted sites."	051 - Software Security, slide 13 051 - Software Security::s13::1	supported (stem gate)	0.57
5	The impact of spyware includes identity theft and compromise of sensitive banking information.	"They can result identity theft and sensitive bank information can be compromised as well."	051 - Software Security, slide 13 051 - Software Security::s13::1	supported (stem gate)	0.67
6	Adware is largely harmless but annoying, generating pop-up advertisements that interrupt other programs.	"Adware as such are quite harmless, although they are annoying. They generate ads in the form of popups and interrupt other programs"	051 - Software Security, slide 13 051 - Software Security::s13::1	supported (stem gate)	0.64

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 051 - Software Security, slide 13

PART A - QUESTION 8

A8a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain how the chain of trust works in the Trusted Computing architecture, with reference to the TPM and the measurement of each stage.

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Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Each stage of the chain measures and validates the following stage, with the resulting measurements recorded in the TPM's Platform Configuration Registers (PCRs).	"Each stage measures and validates the next one"	073 - Trusted Computing, slide 8 073 - Trusted Computing::s08::1	partly supported	0.25 LOW
2	The chain begins at the hardware TPM and proceeds through the software stages (RTM, TPM Software Stack, BIOS, kernel), so that the whole platform ends up verified as trusted.	"The chain starts with the hardware TPM, then software:"	073 - Trusted Computing, slide 8 073 - Trusted Computing::s08::1	partly supported	0.23 LOW

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 073 - Trusted Computing, slide 8

A8b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Compare the two primary system components of the Next Generation Secure Computing Base, namely the Nexus and the Nexus Computing Agent, in terms of their roles and characteristics. Provide six distinct points of comparison.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	The Nexus is described as a special kernel that forms the core of the trusted operating environment.	"Special kernel (core of the trusted operating)"	073 - Trusted Computing, slide 13 073 - Trusted Computing::s13::1	supported (stem gate)	0.62
2	The Nexus aims to isolate normal-mode processes from trusted-mode processes differently in memory.	"Goal: Isolate the process of normal mode and trusted mode differently in memory"	073 - Trusted Computing, slide 13 073 - Trusted Computing::s13::1	supported (stem gate)	0.75
3	The Nexus provides functionality to authenticate and protect data that is entered, stored, communicated, and displayed by means of encryption.	"Functionality: Authenticate and protect data (entered, stored, communicated, and displayed) by data encryption"	073 - Trusted Computing, slide 13 073 - Trusted Computing::s13::1	supported (stem gate)	0.75
4	The Nexus acts as a security kernel that is authenticated on boot.	"Security kernel, authenticated on boot"	073 - Trusted Computing, slide 14 073 - Trusted Computing::s14::1	supported (stem gate)	0.67
5	The Nexus authenticates trusted programs and serves as the application interface to the TPM, while not trusting the OS.	"Authenticates trusted programs"	073 - Trusted Computing, slide 14 073 - Trusted Computing::s14::1	partly supported	0.30 LOW
6	The Nexus Computing Agent is a trusted software component that runs in trusted mode and communicates with the Nexus.	"Trusted software component Runs in trusted mode that communicates with Nexus"	073 - Trusted Computing, slide 15 073 - Trusted Computing::s15::1	supported (stem gate)	0.75

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 073 - Trusted Computing, slide 13; 073 - Trusted Computing, slide 14; 073 - Trusted Computing, slide 15

PART A - QUESTION 9

A9a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

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Explain two password security controls that can be applied to password files and accounts in order to strengthen password-based authentication.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Access control should be applied to the password files so that they are protected from unauthorised reading.	"Access control applied to password files"	022 - Identification and Authentication, slide 12 022 - Identification and Authentication:: s12::1	partly supported	0.40 LOW
2	Accounts should be de-activated after multiple failed login attempts to resist guessing attacks.	"De-activation of accounts after multiple failed logins"	022 - Identification and Authentication, slide 12 022 - Identification and Authentication:: s12::1	supported (stem gate)	0.55

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 022 - Identification and Authentication, slide 12

A9b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain the mechanisms of Kerberos, smartcards and single sign-on as alternative authentication approaches, covering the motivation for single sign-on, the categories of single sign-on solutions, how a Kerberos user login establishes a secret key, and the role of smartcards.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Single sign-on reduces the need for users to memorize passwords.	"Reduce the need to memorize passwords"	022 - Identification and Authentication, slide 16 022 - Identification and Authentication:: s16::1	supported (stem gate)	0.57
2	Single sign-on reduces the time spent typing passwords.	"Reduce the time spent on typing passwords"	022 - Identification and Authentication, slide 16 022 - Identification and Authentication:: s16::1	supported (stem gate)	0.67
3	Single sign-on solutions include local password containers such as KDE Wallet.	"Local password containers (e.g. KDE Wallet)"	022 - Identification and Authentication, slide 16 022 - Identification and Authentication:: s16::1	supported (stem gate)	0.50
4	Single sign-on can be implemented through client-server architectures such as Kerberos and Active Directory.	"Client-server architectures (Kerberos, Active Directory)"	022 - Identification and Authentication, slide 16 022 - Identification and Authentication:: s16::1	supported (stem gate)	0.50
5	In Kerberos login, the hash of the user's password becomes the secret key of the client/user.	"The hash value of his password KU = h(P) becomes a secret key of the client/user."	022 - Identification and Authentication, slide 19 022 - Identification and Authentication:: s19::1	supported (stem gate)	0.75

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No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
6	Smart cards are listed as an alternative user authentication mechanism alongside biometrics and two-factor authentication.	"Alternative user authentication mechanisms are smart cards, biometric methods and two-factor authentication."	022 - Identification and Authentication, slide 24 022 - Identification and Authentication:: s24::1	supported (stem gate)	0.89

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 022 - Identification and Authentication, slide 16; 022 - Identification and Authentication, slide 19; 022 - Identification and Authentication, slide 24

PART A - QUESTION 10

A10a - 2 marks - supported by lecture slides (not lecturer-marked)

Question:

A network administrator is deploying a Network-Based Intrusion Detection System (NIDS) to protect an organization's critical servers. List TWO (2) typical locations where NIDS sensors should be placed.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Sensor placement just inside the firewall	"Just inside the firewall"	062 - Intrusion Detection System, slide 7 062 - Intrusion Detection System:: s07::1	supported (stem gate)	0.60
2	Sensor placement on the DMZ	"On the DMZ"	062 - Intrusion Detection System, slide 7 062 - Intrusion Detection System:: s07::1	supported	0.33 LOW

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Citations: 062 - Intrusion Detection System, slide 7

A10b - 6 marks - supported by lecture slides (not lecturer-marked)

Question:

A company wants to strengthen its network defences before deploying monitoring tools. State SIX (6) preventive network security mechanisms and, for each, briefly describe what it does.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Firewall: filters inbound and outbound network traffic according to predefined security rules.	"Control incoming and outgoing traffic on network with predetermined security policies."	061 - Network Security, slide 3 061 - Network Security::s03::1	supported	0.30 LOW
2	Proxy server: improves data security and network performance, acting as a firewall and web filter while caching data.	"Proxy servers act as a firewall and web filter, provide shared network connections, and cache data to speed up common requests."	061 - Network Security, slide 3 061 - Network Security::s03::1	supported (stem gate)	0.69
3	Network segmentation: creates boundaries between network segments whose assets share a common function, risk or role.	"Boundaries between network segments where computing assets within the segment have a common function, risk or role within an organization."	061 - Network Security, slide 3 061 - Network Security::s03::1	supported (stem gate)	0.75
4	Remote access VPN: gives individual hosts or clients secure remote access to the organization network.	"Provides remote and secure access to an organization network to individual hosts or clients"	061 - Network Security, slide 4 061 - Network Security::s04::1	supported (stem gate)	0.89

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No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
5	Zero Trust Network Access: restricts each user to only the access and permissions needed for their role.	"A user should only has access and permission that required to fulfill the role"	061 - Network Security, slide 4 061 - Network Security::s04::1	supported (stem gate)	0.50
6	IDS/IPS: detects or prevents attacks such as brute force, Denial-of-Service and exploitation of network vulnerabilities.	"To detect or prevent the network security attacks such as brute force attacks, Denial-of-Service attacks and exploits of network vulnerabilities."	061 - Network Security, slide 4 061 - Network Security::s04::1	supported (stem gate)	0.85

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 061 - Network Security, slide 3; 061 - Network Security, slide 4

PART B - QUESTION 1

B1a - 5 marks - supported by lecture slides (not lecturer-marked)

Question:

Explain five ways in which a voluntary professional code of ethics can shape the ethical impact of the technology that computing professionals build and deploy, and note the limits of relying on such a code.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	A code may fail to represent the ethical views of every individual in the group it covers.	"May not reflect the ethical of every member of the group."	071 - Ethics and Computer Security, slide 9 071 - Ethics and Computer Security::s09::1	supported	0.40 LOW
2	Such a code is frequently not enforceable against members.	"Many times it isn't enforceable."	071 - Ethics and Computer Security, slide 9 071 - Ethics and Computer Security::s09::1	supported	0.14 LOW
3	It functions as a code of conduct rather than a legal instrument.	"It's a code of conduct, not a legal document."	071 - Ethics and Computer Security, slide 9 071 - Ethics and Computer Security::s09::1	supported	0.43 LOW
4	Merely describing ethical dimensions of ICT is insufficient; the code must prescribe expected behaviour.	"Simply describing may not be enough when discussing the moral and ethical dimensions of ICT"	071 - Ethics and Computer Security, slide 3 071 - Ethics and Computer Security::s03::1	partly supported	0.30 LOW
5	It reminds users and information specialists that they have definite duties to act within ethical, moral and legal limits.	"Users in general, and information specialists in particular, have definite responsibilities in terms of performing within ethical, moral, and legal constraints"	071 - Ethics and Computer Security, slide 3 071 - Ethics and Computer Security::s03::1	supported (stem gate)	0.64

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 071 - Ethics and Computer Security, slide 3; 071 - Ethics and Computer Security, slide 9

B1b - 5 marks - supported by lecture slides (not lecturer-marked)

Question:

Using the hospital scenario, explain five ethical concerns that the programmer should raise with management regarding the privacy of patients' information and the organization's responsibility for it.

Marking points:

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No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	The organization must safeguard patients' personal information as a matter of privacy.	"Protecting one's personal information"	071 - Ethics and Computer Security, slide 13 071 - Ethics and Computer Security::s13::1	partly supported	0.25 LOW
2	The organization must ensure that stored patient data is accurate and can be verified as genuine.	"Deals with authentication and fidelity of information"	071 - Ethics and Computer Security, slide 13 071 - Ethics and Computer Security::s13::1	partly supported	0.00 LOW
3	The organization must decide who owns the patient information and under what conditions it may be sold or exchanged.	"Deals with who owns information about individuals and how information can be sold and exchanged"	071 - Ethics and Computer Security, slide 13 071 - Ethics and Computer Security::s13::1	supported	0.42 LOW
4	The organization must define which staff have a right to obtain information about patients and how that information may be used.	"Deals with what information a person has the right to obtain about others and how the information can be used"	071 - Ethics and Computer Security, slide 13 071 - Ethics and Computer Security::s13::1	supported	0.36 LOW
5	The organization should recognize that without law and justice, conduct such as unauthorized snooping makes people behave in the worst possible way.	"when separated from law and justice, he is the worst of all"	071 - Ethics and Computer Security, slide 4 071 - Ethics and Computer Security::s04::1	partly supported	0.19 LOW

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 071 - Ethics and Computer Security, slide 13; 071 - Ethics and Computer Security, slide 4

PART B - QUESTION 2

B2a - 5 marks - supported by lecture slides (not lecturer-marked)

Question:

Discuss the goals of access control and the role of a reference monitor in making authorization decisions, then explain how the Chinese Wall model constrains access through its simple security rule and its write property.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	Access control aims to grant access to subjects.	"Goals of access control: Granting access"	023 - Access Control and Authorization, slide 4 023 - Access Control and Authorization::s04::1	supported (stem gate)	0.60
2	Access control also aims to limit access.	"Limiting access"	023 - Access Control and Authorization, slide 4 023 - Access Control and Authorization::s04::1	supported	0.40 LOW

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No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
3	A reference monitor is the component that makes authorization decisions.	"Reference monitor makes authorization decisions."	023 - Access Control and Authorization, slide 4 023 - Access Control and Authorization::s04::1	supported (stem gate)	1.00
4	The Chinese Wall simple security rule permits a subject access to an object only when that object is either within the subject's existing wall (the same company datasets already accessed) or belongs to a completely different conflict of interest class.	"is in the same company datasets as the objects already accessed by s, that is, "within the Wall," or belongs to an entirely different conflict of interest class."	024 - Formal Security Model, slide 25 024 - Formal Security Model::s25::1	supported (stem gate)	0.52
5	The Chinese Wall write property permits write access only when the simple security rule allows it and no readable object in a different company dataset contains unsanitized information.	"Write access is only permitted if:"	024 - Formal Security Model, slide 26 024 - Formal Security Model::s26::1	partly supported	0.21 LOW

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.
Style confidence: LOW - exact examples n<3; widened via topic+marks (any command).
Citations: 023 - Access Control and Authorization, slide 4; 024 - Formal Security Model, slide 25; 024 - Formal Security Model, slide 26

B2b - 5 marks - supported by lecture slides (not lecturer-marked)

Question:

State five key security models and, for each, the distinguishing feature given in the lecture, including the model that identifies conflicts of interest and the model that adds integrity verification.

Marking points:

No.	Point (1 mark)	Evidence quote	Source	Verdict	Overlap
1	The Bell-La Padula (BLP) model is listed as a key security model concerned with access control only.	"Bell-La Padula (BLP) model: access control only"	024 - Formal Security Model, slide 5 024 - Formal Security Model::s05::1	supported (stem gate)	0.78
2	The Biba model is listed as a key security model that performs additional integrity verification.	"Biba model: additional integrity verification"	024 - Formal Security Model, slide 5 024 - Formal Security Model::s05::1	supported (stem gate)	0.71
3	The Chinese Wall model is listed as a key security model for identification of conflicts of interest.	"Chinese Wall model: identification of conflicts of interest"	024 - Formal Security Model, slide 5 024 - Formal Security Model::s05::1	supported (stem gate)	0.86
4	The Take-grant model is listed as a key security model giving a graph-theoretical interpretation of an access matrix.	"Take-grant model: graph-theoretical interpretation of an access matrix"	024 - Formal Security Model, slide 5 024 - Formal Security Model::s05::1	supported (stem gate)	0.89
5	The Chinese Wall policy, proposed by Brewer and Nash in 1989, is described as addressing conflicts of interest.	"Brewer and Nash (1989) proposed a policy called the Chinese Wall Policy that addresses such conflicts of interest."	024 - Formal Security Model, slide 21 024 - Formal Security Model::s21::1	supported (stem gate)	1.00

Confidence: supported by lecture slides (not lecturer-marked) - each point carries a verbatim quote from the cited lecture slide.

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Citations: 024 - Formal Security Model, slide 21; 024 - Formal Security Model, slide 5

End of marking scheme.

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